

(a) From Eq. (7.59), the refraction index is

$$n(\omega) = \sqrt{\epsilon(\omega)/\epsilon_0} = \sqrt{\omega^2 - \omega_p^2} / \omega.$$

The total reflection angle is $i_s = \arcsin(n(\omega)) = \arcsin\left(\sqrt{1 - \frac{\omega_p^2}{\omega^2}}\right)$

(b) In this case, the total reflection angle is

$$\sin i_s = \frac{5}{\sqrt{34}}, \text{ or } \frac{5}{\sqrt{34}} = \sqrt{1 - \frac{Nz e^2}{\omega^2 \epsilon_0 m}}$$

Here, $\omega = \frac{2\pi c}{\lambda}$, and

$$Nz = \frac{q}{34} \frac{\omega^2 \epsilon_0 m}{e^2} = \frac{36\pi^2 c^2}{34\lambda^2} \frac{\epsilon_0 m}{e^2}$$

$$= \frac{36 \times \pi^2 \times (299792458)^2}{34 \times (21)^2} \times \frac{8.854 \times 10^{-12} \times 9.1093837 \times 10^{-31}}{(1.602176634 \times 10^{-19})^2}$$

$$= 6.69166 \times 10^{11}$$